

Algorithm: CustomCNN with CustomMaxPool2D, CBAM, and Grad-CAM

1: Define CustomMaxPool2D:

```
if S = None → S ← K
if P > 0 → Pad I by P
Extract patches using Unfold(K, S)
O ← max(Flattened patches)
return O
```

2: Define CBAM:

```
channel_att ← sigmoid(FC(avg_pool(I)) + FC(max_pool(I)))
I ← I * channel_att
spatial_att ← sigmoid(conv(concat(avg(I), max(I))))
I ← I * spatial_att
return I
```

3: Define CustomCNN:

```
For each block (1–4):
    I ← Conv2d → BatchNorm → ReLU
    I ← CustomMaxPool2D(I)
    I ← CBAM(I)
    Apply Dropout
I ← AdaptiveMaxPool2d → Flatten → FC → ReLU → FC(num_classes)
return I
```

4: Define Grad-CAM:

```
output ← model(I)
if class_idx = None → class_idx ← argmax(output)
one_hot[class_idx] ← 1
output.backward(gradient=one_hot)
weights ← mean(gradients)
H ← relu(sum(weights * feature_maps))
Normalize H
return H, class_idx
```

5: Forward Pass:

```
preds ← CustomCNN(I)
heatmap, cls ← Grad-CAM(preds)
return preds, heatmap
```